

Submission Summary

Conference Name

IEEE Chandigarh Subsection International Conference

Track Name

Artificial Intelligence and Data Science

Paper ID

1634

Paper Title

A Comparative Study of Swin Transformer and ResNet50 for Automated Detection of Age-Related Macular Degeneration Using Fundus Images

Abstract

Age-related macular degeneration (AMD) is a major reason for the irreversible loss of vision in the central portion of the visual field and is a significant problem in people over the age of 75 years worldwide. The problem of early diagnosis is still a big issue due to the interpretation of retinal images and the lack of availability of expert ophthalmologists. To handle the problem of data imbalance and improve the generalization of the model, the Synthetic Minority Oversampling Technique (SMOTE) is used in this study to generate synthetic images of the minority class, i.e., age-related macular degeneration. The study focuses on the application of deep learning models, specifically the Swin Transformer and ResNet50, for the detection of age-related macular degeneration from retinal images. The Swin Transformer is based on the hierarchical window-based self-attention mechanism, which is effective in extracting global features from the images, and ResNet50 is a deep convolutional neural network that is effective in extracting local spatial features and is computationally efficient even on small datasets.

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Submission Files

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Submission Questions Response**1. Corresponding author Email ID**

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3. Author's Information

I confirm that I have listed the title of the paper and the names, email addresses and affiliations of all contributed authors in the submission system that match the order as given in the submitted paper

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4. Co-author's Consent

I confirm that I have discussed about this submission with all co-authors and authors have approved the paper and agree with its submission to this conference.

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I confirm that neither the paper nor any parts of its content are currently under consideration or published in another conference/journal and the paper is free from plagiarism

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